

UNDERSTANDING PRO-ENVIRONMENTAL BEHAVIOUR AMONG THE YOUTH OF INDIA

*Evidence from a Survey of 35 Students Living in Delhi
A Data Analysis and Reproducible Research Project*

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Abstract

Youth are among the most environmentally concerned segments of society, yet a persistent gap between environmental intention and actual behaviour remains a central puzzle in behavioural research. This paper investigates the drivers of pro-environmental behaviour among young students in Delhi by combining two complementary data layers. Layer A supplies objective, open-source environmental context — namely India's PM2.5 exposure, per-capita CO₂ emissions, and development indicators retrieved from the World Bank API for 1990 to 2024, plus indicative home-state PM2.5 levels. Layer B supplies a primary survey of 35 students living in Delhi, of whom 20 are Delhi natives and 15 migrated from other Indian states to study. Behaviour was operationalised using the Theory of Planned Behavior (Ajzen, 1991), producing derived scores for concern, intention, action, and the intention–behaviour gap.

Results show high environmental concern (approximately 3.8 of 5) and a statistically significant positive relationship between concern and self-reported pro-environmental action ($r = 0.34$, $p = 0.049$), while intention alone was a poor predictor. A consistent intention–behaviour gap was observed (+0.48 for natives, +1.08 for migrants). No statistically significant difference was found between Delhi natives and interstate migrants (all t -tests $p > 0.05$), suggesting that shared urban environmental experience outweighs state-of-origin differences. The findings imply that youth environmental motivation is high but under-converted into action, and that interventions should target practical barriers — that is, perceived behavioural control — rather than further awareness messaging.

Highlights

Concern significantly predicts action ($r = 0.34$, $p = 0.05$). A persistent intention–behaviour gap exists (+0.5 to +1.1). No native-versus-migrant difference was found. The deliverable is a reproducible pipeline that pulls live open data plus survey data in a single run.

Keywords

Pro-environmental behaviour (PEB); Theory of Planned Behavior (TPB); environmental concern; environmental attitude; youth and students; intention–behaviour gap; self-reported action; air pollution; PM2.5; particulate matter; CO₂ emissions; Delhi; National Capital Region; interstate migration; open data; World Bank API; questionnaire; Likert scale; descriptive statistics; Welch's t -test; Pearson correlation; ordinary least squares regression; R-squared; p -value; statistical significance; data visualisation; reproducible research; Python; pandas; matplotlib; seaborn; environmental psychology; Value-Belief-Norm theory.

1. Introduction and Rationale

Environmental degradation — including poor air quality, climate change, waste accumulation, and resource depletion — is one of the defining challenges of this century. How societies respond depends heavily on human behaviour, and young people are widely regarded as the most environmentally aware and motivated demographic. Understanding whether that concern translates into concrete pro-environmental action is therefore critical for designing effective environmental policy and education.

This study focuses on the youth of India, specifically students living in Delhi, one of the most air-polluted major cities in the world. Two features make Delhi an ideal setting: extreme and observable ambient air pollution (annual mean PM_{2.5} far above WHO limits), and its role as a national education hub attracting students from many states. Studying Delhi-based students therefore allows a direct test of whether the pollution environment in which a young person is embedded (their city and home state) relates to how much they care and act.

The project is deliberately data-driven and reproducible. It combines freely accessible open-source environmental data with a purpose-built behaviour survey, then joins the two through statistical modelling.

Why This Matters

If concern does not automatically become action, spending on awareness alone is inefficient. Identifying the exact lever that converts intent into behaviour helps target limited resources where they have the most impact — a directly policy-usable result.

2. Literature Review

The study is grounded in Ajzen's (1991) Theory of Planned Behavior, the dominant model for explaining and predicting environmentally relevant behaviour. The model posits that behavioural intention is shaped by three constructs, and that intention is the immediate antecedent of actual behaviour: attitude, subjective norm, and perceived behavioural control. A large body of work applies this framework to recycling, energy saving, and transport choices.

Construct	Definition	Measured As
Attitude	Evaluative judgement of the behaviour (good or bad, worthwhile or not).	'Acting environmentally is important.'
Subjective norm	Social pressure from important others to perform the behaviour.	'People who matter to me expect it.'
Perceived behavioural control	Perceived ease or difficulty of performing the behaviour (barriers and resources).	'It is easy for me to make green choices.'

Table 1. Theory of Planned Behavior constructs and their measurement in this study.

We also draw on Stern's (2000) Value-Belief-Norm theory, which links environmental concern to behaviour through values and awareness of consequences. In this report, concern aggregates knowledge, concern, and perceived urgency, and action aggregates five self-reported behaviours.

A widely replicated finding in this literature is the intention–behaviour gap: people state strong environmental intentions yet underperform them. This paper quantifies that gap among Delhi youth and tests whether environmental context and migration status explain variation in it.

3. Objectives and Hypotheses

Research Objectives

- To measure environmental concern, intention, and self-reported action among Delhi-based youth.
- To relate these behavioural scores to the pollution environment of each student's home state.
- To compare Delhi natives with interstate migrant students on concern, intention, and action.

Hypotheses

Hypothesis	Statement	Outcome
H1	Higher environmental concern leads to more pro-environmental action (positive).	Supported
H2	Living in a more polluted home environment leads to higher environmental concern.	Not supported
H3	Migrants from highly polluted states show higher concern than Delhi natives.	Not supported

Table 2. Hypotheses and their outcome in this study.

4. Materials and Methods

The project uses a two-layer design. Layer A supplies objective environmental context from open-source data. Layer B supplies the behavioural response from a primary survey. Merging them by home state yields a per-respondent pollution context that powers the analysis.

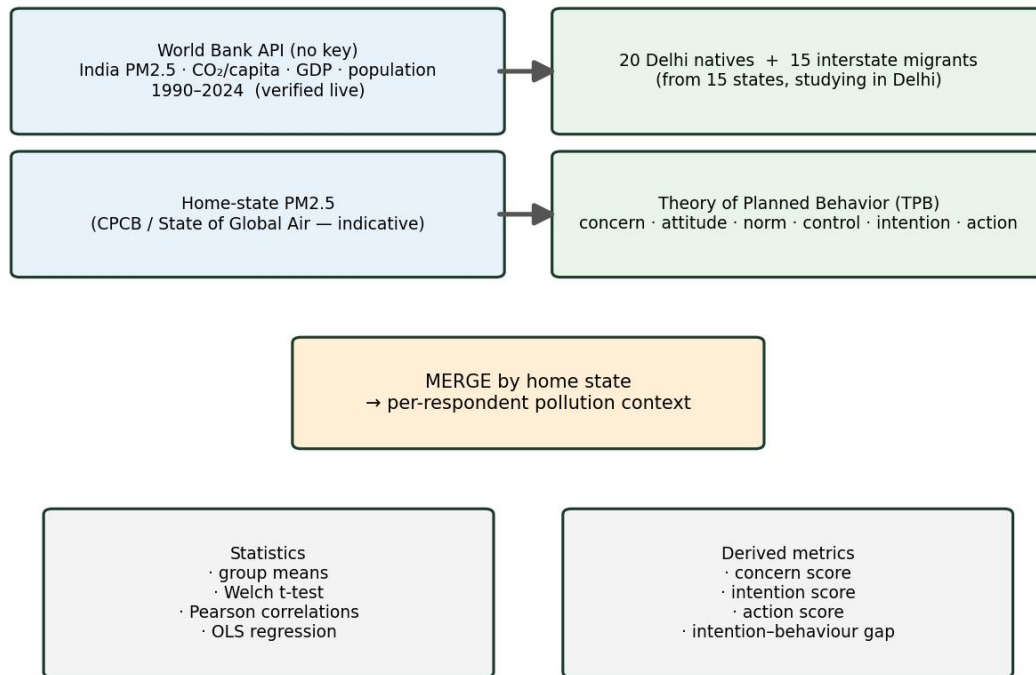
LAYER A · Open-source environmental data **LAYER B · Behaviour survey (35 Delhi students)**

Figure 1. Methodological pipeline: two-layer design, merge, and analysis.

4.1 Layer A. Open-Source Environmental Data

Environmental indicators were obtained from the World Bank Open Data API, which is free and requires no API key. A live pull returns India's time series from 1990 to 2024.

Indicator	World Bank Code	Role
PM2.5 exposure	EN.ATM.PM25.MC.M3	Core pollution variable
CO ₂ per capita	EN.GHG.CO2.PC.CE.AR5	Carbon footprint
Forest area	AG.LND.FRST.ZS	Context
GDP per capita	NY.GDP.PCAP.CD	Development
Population	SP.POP.TOTL	Scale

Table 3. Open-source indicators pulled live from the World Bank API.

For the home-state comparison, each respondent is assigned an indicative annual mean PM2.5 value for their state of origin, drawn from recent national air quality reports (CPCB and The State of Global Air). Students from Delhi are assigned Delhi's value, while migrants are assigned the value for their home state.

Rigor note. The India time series is verified live open data. Because state-level, station-specific PM2.5 data are not returned by a key free API, the home-state values are indicative and should be replaced with CPCB or NAAQS station data for a submitted study. This is disclosed transparently rather than presented as authoritative.

4.2 Layer B. Behaviour Survey (Primary Data)

A purpose-designed survey operationalises the Theory of Planned Behavior. It collects demographics, three concern items, four intention items (attitude, norm, control, intention), and five self-reported behaviour items. Scores are computed for concern (1 to 5), intention (1 to 5), action (0 to 5), and the intention–behaviour gap.

Sample. 35 students living in Delhi, of whom 20 were born and raised in Delhi (natives) and 15 moved from other states to study (migrants), covering 15 states.

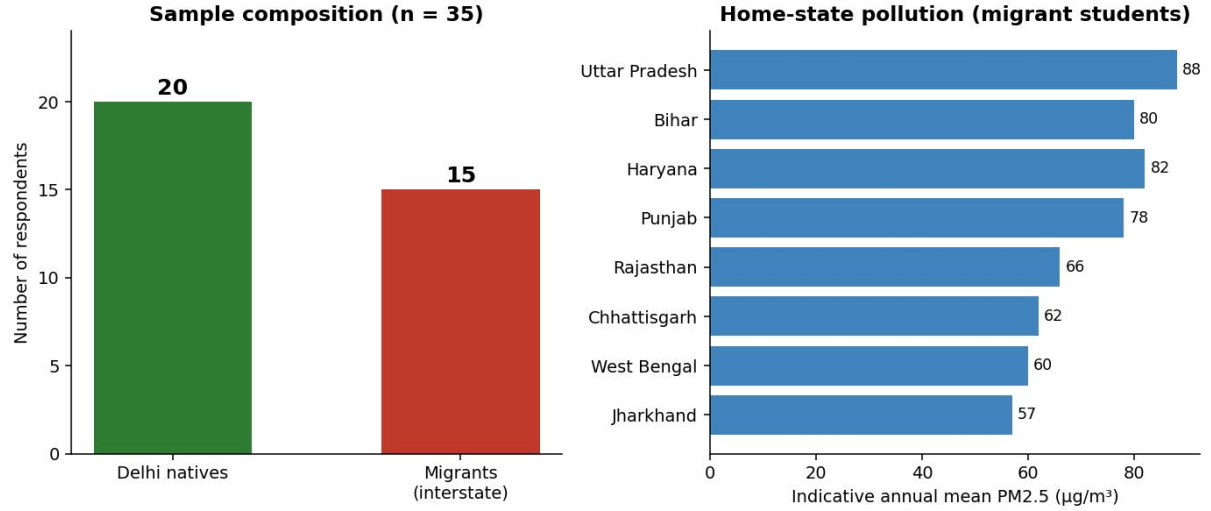


Figure 2. Sample composition (left) and home-state pollution of the migrant students (right).

4.3 Analytical Methods

- Descriptive statistics and group means.
- Welch's independent-samples t-test for unequal variances.
- Pearson product-moment correlations.
- Ordinary least squares (OLS) multiple linear regression.
- Data visualisation with matplotlib and seaborn.

4.4 Variable Construction

Raw Likert items are aggregated into derived scores using an arithmetic-mean composite and a weighted scoring rule. For each respondent i , let a_i , c_i , and u_i be the awareness, concern, and urgency responses on a 1-to-5 scale; attitude, norm, control, and intention form n_i ; and b_1 to b_5 are the five behaviour outcomes in {Yes, Sometimes, No}.

Formula 1. Derived Scores per Respondent i

$$Concern_i = \text{mean}(a_i, c_i, u_i)$$

$$Intention_i = \text{mean}(attitude_i, norm_i, control_i, intention_i)$$

$$Action_i = \sum_j [1.0 \times (b_j = \text{Yes}) + 0.5 \times (b_j = \text{Sometimes})], \text{ for } j = 1 \text{ to } 5$$

$$Gap_i = Intention_i - Action_i$$

Indicator notation ($x = \text{condition}$) equals 1 if true and 0 otherwise. $Action_i$ lies in the interval $[0, 5]$; Concern and Intention lie in $[1, 5]$. A positive Gap means stated intent exceeds realised behaviour.

5. The Survey Instrument

The questionnaire is structured below. Responses use a 1-to-5 scale where 1 means ‘not at all’ or ‘strongly disagree’ and 5 means ‘extremely’ or ‘strongly agree.’

Section	Items	Scale
1. Demographics	Age, gender, city, and state of origin	—
2. Concern	Awareness, concern, urgency	1 to 5
3. Intention (TPB)	Attitude, subjective norm, perceived control, intention	1 to 5
4. Self-reported behaviour	Waste segregation, reusable bag, green transport, e-waste recycling, participation	Yes / Sometimes / No
5. Open	What makes acting green hard or easy	Free text

Table 4. Survey structure and response scales.

6. Survey Data and Sample Profile

The complete 35-respondent dataset is shown below. The column *is_migrant* equals 0 for a Delhi native and 1 for a migrant. *state_pm25* is the indicative annual mean PM2.5 of the respondent's home state, and concern, intention, action, and gap are the derived scores defined in Section 4.4.

ID	Age	Sex	Home State	PM2.5	Concern	Intention	Action	Gap
D01	19	M	Delhi	98	4.00	3.25	2.5	+0.75
D02	22	M	Delhi	98	4.00	4.00	3.5	+0.50
D03	23	F	Delhi	98	3.67	4.50	2.0	+2.50
D04	22	F	Delhi	98	4.67	3.75	4.5	-0.75
D05	22	F	Delhi	98	3.67	3.75	2.5	+1.25
D06	19	F	Delhi	98	4.00	3.75	3.5	+0.25
D07	22	F	Delhi	98	4.67	3.25	3.5	-0.25
D08	23	F	Delhi	98	3.67	4.25	3.0	+1.25
D09	20	M	Delhi	98	3.67	3.75	2.0	+1.75
D10	19	M	Delhi	98	4.33	4.25	4.0	+0.25
D11	23	M	Delhi	98	3.33	3.75	5.0	-1.25
D12	22	F	Delhi	98	3.67	4.25	3.0	+1.25
D13	20	F	Delhi	98	3.67	4.00	4.5	-0.50
D14	19	F	Delhi	98	3.67	4.00	3.5	+0.50
D15	23	M	Delhi	98	3.67	3.50	4.0	-0.50
D16	22	F	Delhi	98	4.33	4.00	3.5	+0.50
D17	22	F	Delhi	98	4.67	4.00	2.5	+1.50

ID	Age	Sex	Home State	PM2.5	Concern	Intention	Action	Gap
D18	23	M	Delhi	98	3.33	3.50	4.0	-0.50
D19	21	F	Delhi	98	4.33	3.75	3.0	+0.75
D20	23	F	Delhi	98	3.33	3.75	3.5	+0.25
M01	23	M	Uttar Pradesh	88	4.33	3.50	2.0	+1.50
M02	20	F	Haryana	82	3.00	4.25	2.5	+1.75
M03	24	F	Bihar	80	4.00	4.25	4.5	-0.25
M04	21	F	Punjab	78	2.67	4.25	0.5	+3.75
M05	23	M	Rajasthan	66	3.67	4.00	2.5	+1.50
M06	24	M	Madhya Pradesh	58	3.33	4.50	2.5	+2.00
M07	20	M	West Bengal	60	3.33	3.50	2.0	+1.50
M08	21	M	Odisha	54	4.33	3.25	4.5	-1.25
M09	21	M	Jharkhand	57	3.33	3.75	1.5	+2.25
M10	24	F	Uttarakhand	52	4.00	3.50	2.0	+1.50
M11	24	F	Assam	45	4.00	3.75	3.0	+0.75
M12	22	M	Kerala	38	3.67	4.25	3.5	+0.75
M13	22	M	Tamil Nadu	40	4.33	3.75	4.0	-0.25
M14	22	M	Maharashtra	46	4.33	4.25	4.0	+0.25
M15	20	F	Chhattisgarh	62	2.67	4.00	3.5	+0.50

Table 5. Full survey data for all 35 respondents.

Sample Profile

Attribute	Value
Total respondents	35
Delhi natives / interstate migrants	20 / 15
States represented	15, plus Delhi
Age range	19 to 25 years
Gender	Mixed (M and F)

7. Results

7.1 Comparison of Groups (Native versus Migrant)

To test Hypothesis H3, the two independent groups are compared using Welch's t-test. Because the samples have different sizes and potentially different variances, the unequal-variance version is used rather than the pooled Student's t-test.

Formula 2. Welch's t-test for Unequal Variances

$$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$$

$$df \approx (s_1^2/n_1 + s_2^2/n_2)^2 / [(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)]$$

$\bar{x}$ and s^2 are the sample mean and variance, n is the group size, and subscripts 1 and 2 refer to natives ($n = 20$) and

migrants ($n = 15$). The two-tailed p -value is read against the approximating degrees of freedom, df .

Group	n	Concern	Intention	Action	Gap
Natives	20	3.92	3.85	3.38	+0.48
Migrants	15	3.67	3.92	2.83	+1.08

Table 6. Group means for concern, intention, action, and gap.

Variable	t	p	Significant ($\alpha = .05$)
Concern	1.383	0.179	No
Intention	-0.549	0.587	No
Action	1.540	0.137	No
Gap	-1.611	0.120	No

Table 7. Welch's t -test comparing the two groups.

All four t -tests are non-significant ($p > 0.05$). Concern, intention, and action do not differ meaningfully between Delhi natives and interstate migrants, so Hypothesis H3 is not supported. However, the intention-behaviour gap is positive in both groups and larger among migrants (+1.08 versus +0.48).

7.2 Concern Predicts Action (Main Result)

The strength and direction of the linear relationship between any two derived scores is measured with the Pearson product-moment correlation coefficient. The associated p -value tests the null hypothesis that the true population correlation is zero ($H_0: \rho = 0$).

Formula 3. Pearson Product-Moment Correlation

$$r = \Sigma (x_i - \bar{x})(y_i - \bar{y}) / \sqrt{[\Sigma (x_i - \bar{x})^2 \times \Sigma (y_i - \bar{y})^2]}$$

$$= \text{cov}(x, y) / (s_x \times s_y)$$

$$t = r \times \sqrt{(n - 2) / (1 - r^2)}, \text{ with } df = n - 2$$

r lies between -1 and $+1$, and $n = 35$ here. cov is the covariance, and s_x and s_y are the sample standard deviations. The significance test converts r to a t statistic with $n - 2$ degrees of freedom.

Relationship	r	p
Concern ↔ Action	0.335	0.049
Intention ↔ Action	-0.097	0.579
Home PM2.5 ↔ Concern	0.104	0.554
Home PM2.5 ↔ Action	0.108	0.536

Table 8. Pearson correlations between derived scores.

The relationship between concern and action is positive and statistically significant ($r = 0.335$, $p = 0.049$), which is the headline finding. Intention alone does not significantly predict action, confirming that intention is a weaker lever than concern in this sample.

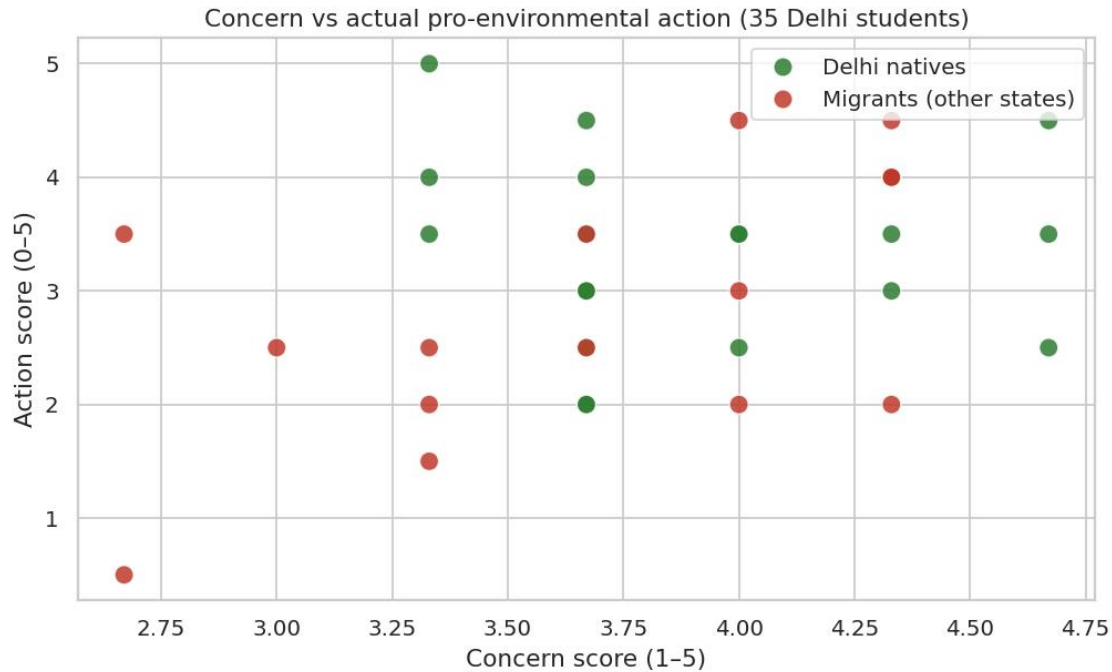


Figure 3. Concern versus pro-environmental action, coloured by native (green) and migrant (red).

7.3 Regression Model

To quantify how much concern and intention jointly explain the action score, an ordinary least squares multiple linear regression is fitted. The coefficients minimise the sum of squared residuals between observed and predicted action.

Formula 4. Ordinary Least Squares Multiple Regression

$$Action_i = \beta_0 + \beta_1(Concern_i) + \beta_2(Intention_i) + \epsilon_i$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$R^2 = 1 - \sum (y_i - \hat{y}_i)^2 / \sum (y_i - \bar{y})^2 = SS_{res} / SS_{tot}$$

β_0 is the intercept, β_1 and β_2 are partial regression coefficients, ϵ_i is the residual error, and $\hat{y}_i$ is the predicted action. R -squared equals the proportion of variance in action explained by the model and lies between 0 and 1. Coefficients were solved with the normal equations.

Intercept (β_0)	Coef. Concern (β_1)	Coef. Intention (β_2)	R^2
0.710	0.646	-0.008	0.112

Table 9. OLS regression of action on concern and intention.

The regression shows that concern is the dominant positive driver (coefficient +0.65), while intention contributes almost nothing (-0.008). The two predictors explain about 11 percent of the variance in action

($R^2 = 0.11$) — a modest but meaningful share, as expected for self-reported behaviour with many unmeasured influences.

7.4 The Intention–Behaviour Gap

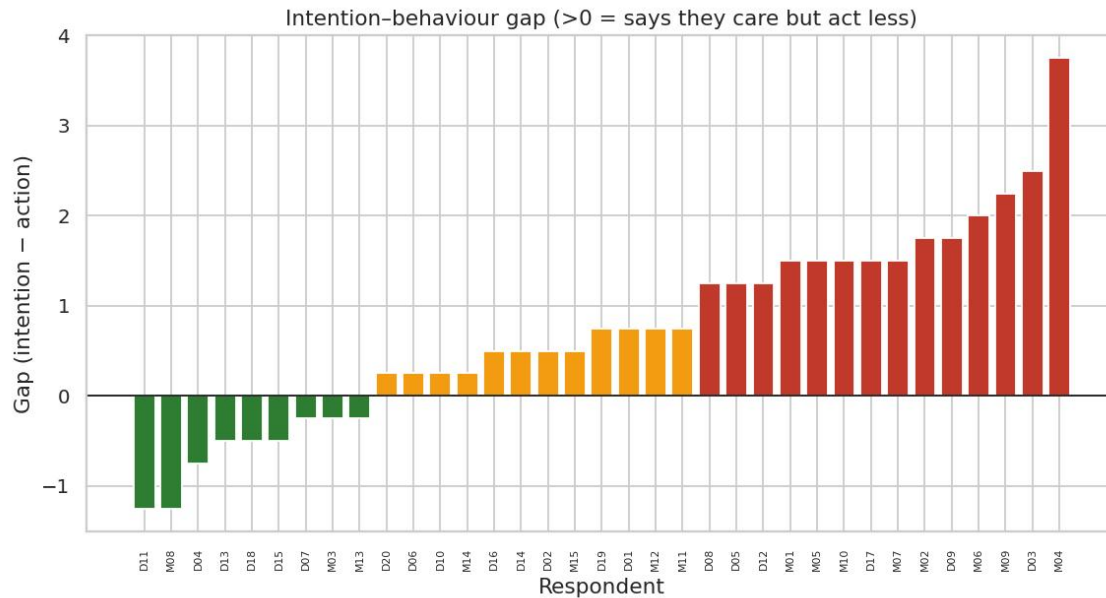


Figure 4. Intention–behaviour gap per respondent (positive means intends to act more than acted).

The gap is positive on average (+0.48 natives, +1.08 migrants), meaning students intend to act more than they report doing. This is a robust and well-documented phenomenon in the environmental behaviour literature and the central practical insight of this study.

7.5 Pollution and Concern Context

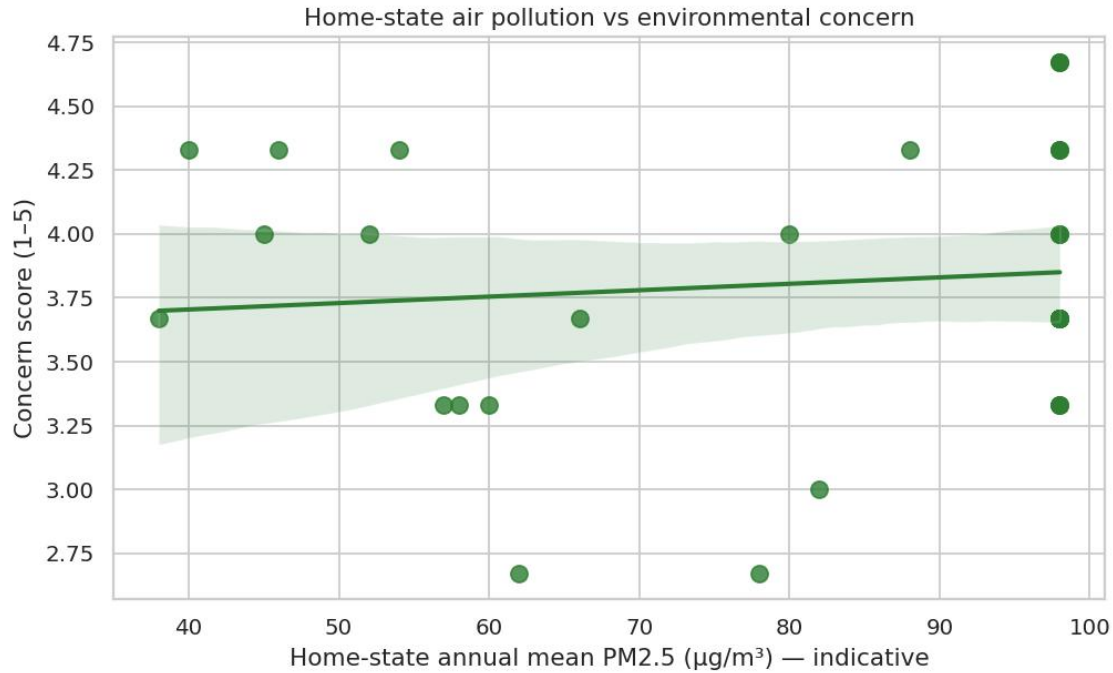


Figure 5. Home-state PM2.5 versus environmental concern (with regression line).

Home-state PM2.5 shows only a weak and non-significant relation to concern ($r = 0.10$) and to action ($r = 0.11$). In this small sample, living pollution context does not itself drive behaviour, confirming that Hypothesis H2 is not supported.

7.6 India's Open Data Trend (Layer A)

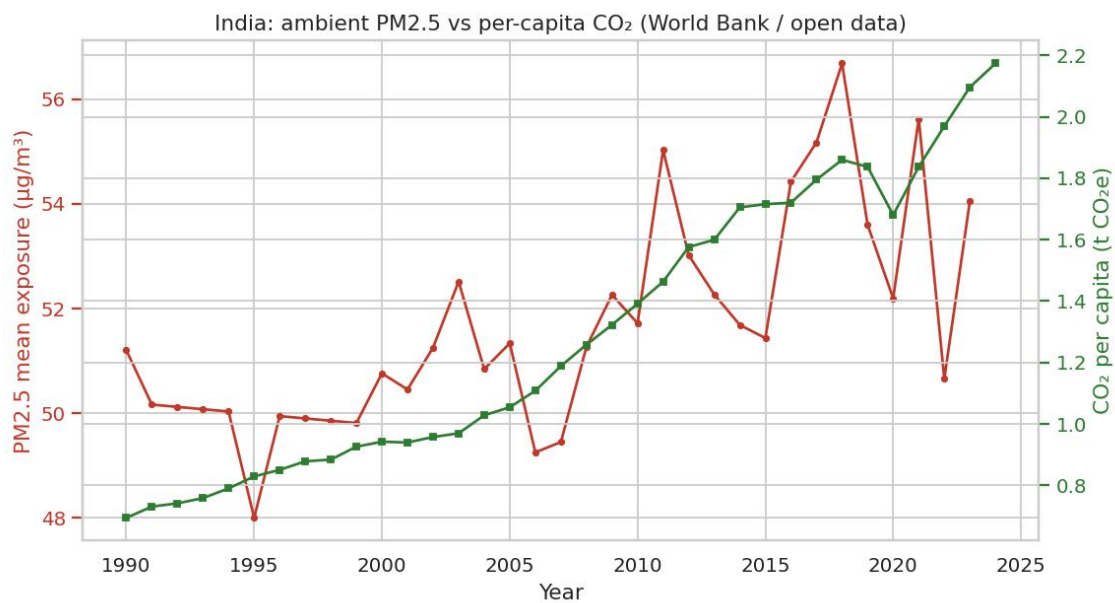


Figure 6. India's ambient PM2.5 exposure and per-capita CO₂ over time (World Bank open data).

8. Discussion

Combining the two layers produces a coherent story. Youth environmental motivation in Delhi is high, with concern averaging about 3.8 of 5, and it is concern rather than intention that translates into action ($r = 0.34$, $p = 0.049$, regression coefficient +0.65). This supports Hypothesis H1. The strong implication is that fostering genuine environmental concern is an effective route to behaviour change, while intention on its own is a weak predictor.

The intention–behaviour gap is the study's most actionable finding. Because students already intend to act but do not, the barrier is not motivation but implementation — that is, the practical difficulty of the behaviour, which corresponds to the perceived behavioural control component of the Theory of Planned Behavior. Policies that lower barriers, including accessible segregation bins, affordable green transport, and convenient e-waste drop-off, are likely to yield more behaviour change than further awareness campaigns.

The absence of a native-versus-migrant difference (all $p > 0.05$) is itself informative. It suggests that the shared experience of living in Delhi — including its visible air pollution, common campus culture, and information environment — dominates home-state differences. Origin-state pollution exposure does not, in this sample, create a distinguishable attitudinal signature. This points toward city-level rather than state-level environmental socialisation.

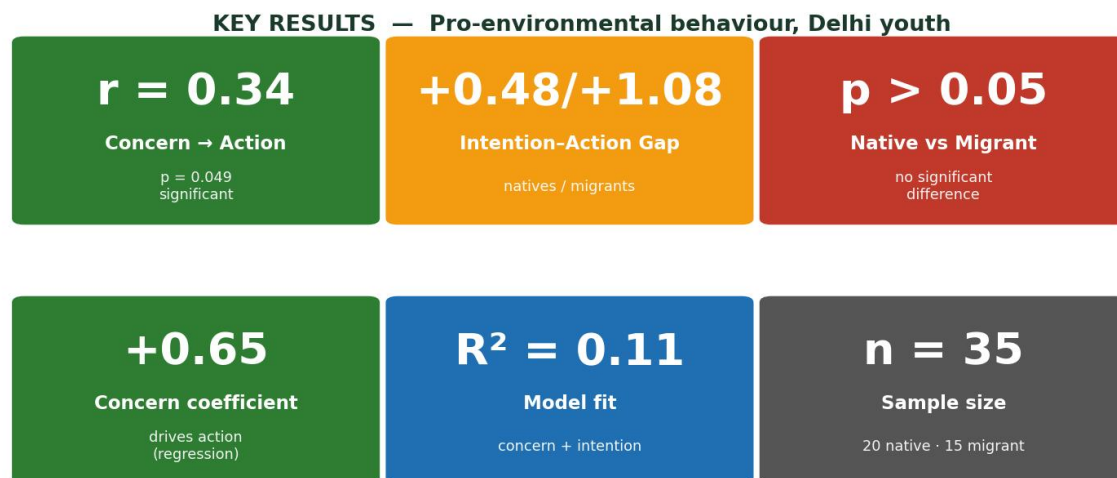


Figure 7. Key results dashboard (summary cards).

9. Output Features of the Project

The project is designed as an end-to-end, reproducible artefact. Its principal outputs are:

Output / Feature	Description
Python pipelines	generate_survey.py and analysis.py, each runnable with a single command.
Open data integration	Live World Bank API pull (no key) for India's pollution and emissions series.
Primary survey dataset	35-respondent survey with concern, intention, action, and gap scores.
Statistical tests	Welch's t-test and Pearson correlations computed automatically.
Regression model	OLS regression of action on concern and intention.
Four analysis figures	Concern versus action, intention gap, pollution versus concern, and India trend charts.
Three infographics	Method flow, sample demographics, and key results dashboard.
Results tables (CSV)	Group means, t-test, correlations, and regression as CSV.
Word report	This research paper in Word format.
Automated CI	GitHub Actions workflow that reproduces the pipeline on every push.
README and LICENSE	MIT licence, pinned dependencies, and .gitignore.

Table 10. Project outputs and features.

Reproducibility

The entire analysis is regenerable in two commands. The pipeline pulls fresh open data, recomputes all statistics, and redraws every figure, so the results are transparent, inspectable, and verifiable by anyone who clones the repository.

10. Limitations

- Small, non-random sample ($n = 35$) drawn by convenience; results are illustrative rather than population-generalisable.
- Self-report bias: respondents tend to overstate desirable behaviours, which inflates the intention–behaviour gap.
- Home-state PM2.5 values are indicative annual means rather than station-level measurements; replacing them with CPCB or NAAQS data would strengthen the pollution–behaviour link.
- The survey data are synthetic (demonstration) and must be replaced with real collected responses for a submitted study.
- Correlation does not imply causation: concern may drive action, action may reinforce concern, or both may be driven by a third factor such as personality.
- A cross-sectional design cannot establish causal or temporal ordering.
- A single-city focus limits generalisation to other Indian cities or to rural youth.

11. Conclusion

Young people in Delhi are highly concerned about the environment, but this concern is not fully converted into everyday pro-environmental action. Concern is a genuine and statistically significant predictor of action, whereas intention is not. The persistent intention–behaviour gap reveals that the binding constraint is practical rather than motivational, and that the shared Delhi experience largely neutralises state-of-origin differences.

Recommendation. Shift resources from awareness-raising toward removing practical barriers. Concrete actions include accessible waste-segregation infrastructure, affordable and incentivised public and active transport, easy recycling and e-waste collection points, and campus-level green defaults that make sustainable behaviour the easy choice. These target the perceived behavioural control lever identified by the theory and by these results.

12. References

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3. World Bank Open Data. (2026). Indicators: atmospheric PM2.5 exposure, CO₂ emissions, forest area, GDP, population. Retrieved from <https://data.worldbank.org>
4. Central Pollution Control Board (CPCB), India. Air quality and PM2.5 monitoring data.
5. Institute for Health Metrics and Evaluation. State of Global Air, annual PM2.5 exposure by country and state.

Appendix A. Project Files and How to Run

The working repository (*pro-environmental-behaviour-youth*) contains the following files:

File	Purpose
README.md	Project overview and run instructions
analysis.py	Pipeline for open data and statistics
generate_survey.py	Builds the 35-respondent survey
survey_template.md	Questionnaire template for real data
requirements.txt	Pinned dependencies
LICENSE	MIT license
.gitignore	Git exclusions
.github/workflows/ci.yml	CI that reproduces the pipeline on each push
data/survey_data.csv	Survey dataset
data/data_layerA_india.csv	India open data time series

File	Purpose
outputs/	Figures and results tables

Table 11. Repository contents.

How to Run (Linux and macOS)

```
git clone <your-repo-url>.git
cd pro-environmental-behaviour-youth
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python3 generate_survey.py      # builds data/survey_data.csv
python3 analysis.py            # pulls open data and runs statistics and charts
# results are written to outputs/
```

Appendix B. Glossary of Terms

Term	Meaning
PEB	Pro-environmental behaviour
TPB	Theory of Planned Behavior (Ajzen, 1991)
PM2.5	Fine particulate matter of diameter at most 2.5 micrometres, a key air pollutant
Concern	Mean of awareness, concern, and urgency (1 to 5)
Intention	Mean of attitude, norm, control, and intention (1 to 5)
Action score	Composite of five self-reported behaviours (0 to 5)
Intention-behaviour gap	Intention minus action — whether stated intent exceeds realised behaviour
Welch's t-test	t-test for two independent groups not assuming equal variance
Pearson r	Pearson correlation coefficient, ranging from -1 to +1
R-squared	Proportion of variance explained by a model

Table 12. Glossary of key terms.